Problem Set 6

POLS W4732

Fall 2026

Due November 20 before the TA section

Artificial-intelligence policy. Unless an exercise explicitly states otherwise, students may not use generative-AI systems (including ChatGPT, Claude, Gemini, Copilot, or AI-generated search answers) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, typesetting software, the teaching team, and collaboration within the registered problem-set group are permitted. Any approved computational tool must be disclosed. Every group member is responsible for understanding and being able to explain the entire submission.

Exercise 1 (Media bias and career concerns)**.** *An incumbent's first-period performance is*

$$\pi_1 = \theta_I + e_1 + \varepsilon_1,$$

where $\theta_I \sim N(m, \sigma_\theta^2)$ and effort costs $e_1^2/2$. Reelection is worth B. The voter compares the incumbent's posterior expected ability with a challenger of expected ability m and correctly anticipates equilibrium effort e_1^a.

1. *Environment A: $\varepsilon_1 \sim N(b, \sigma_\varepsilon^2)$ and b is common knowledge. Derive the posterior mean, reelection cutoff, winning probability, and effort first-order condition.*
2. *Environment B: the true bias is b, but the voter believes it is $\widehat{b} \neq b$. Repeat the analysis.*
3. *Environment C: the mean bias is known but its realization contains an additional independent normal component with variance σ_b^2. Repeat the analysis.*
4. *State the exact condition under which additive bias cancels from equilibrium effort. Evaluate the claim that favorable coverage must reduce effort.*

Throughout the updating calculation the observed first-period shock is ε_1 (not ε_2).

Exercise 2 (Electoral advantage and information precision)**.** *In the baseline two-period career-concern model, the voter obtains an additional payoff κ when the incumbent is reelected. Ability is $N(m, \sigma_\theta^2)$, performance noise is $N(0, \sigma_\varepsilon^2)$, office benefit is B, and effort cost is $c(e)$.*

1. *Derive the voter's posterior and show that the reelection cutoff is*

$$\pi^* = e_1^a + m - \frac{\kappa}{\lambda}, \qquad \lambda = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2}.$$

2. *Derive the equilibrium effort condition and state the second-order qualification.*
3. *For* $c(e) = e^2/2$*, test the assertion that more precise information always raises effort. Derive the sign rather than relying on verbal intuition.*
4. *Explain separately the accountability and electoral-security effects. Give one explicit modification of the baseline assumptions under which the precision comparative static could reverse, and state the condition needed.*